

Supplement for RDD results

2026-09-14 15:41

Abstract

Quarto document renders the outputs for the full set of RDD results (sensitivity check etc). Results are also saved in /paper-outputs as a .tex fragments so they can be pulled into overleaf directly

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1 Topline

- robust effect for authors a 3rd note (`have_another_note = note_counts[(note_counts >= 3)]`)
- with authors a 2nd note (`have_another_note = note_counts[(note_counts >= 2)]`)
- – robust rd not sig, donut RDD sig for 2 “larger holed donuts”, ordinary OLS is sig with 0.08 threshold,
- RDD is not sharp

2 Setup & Data

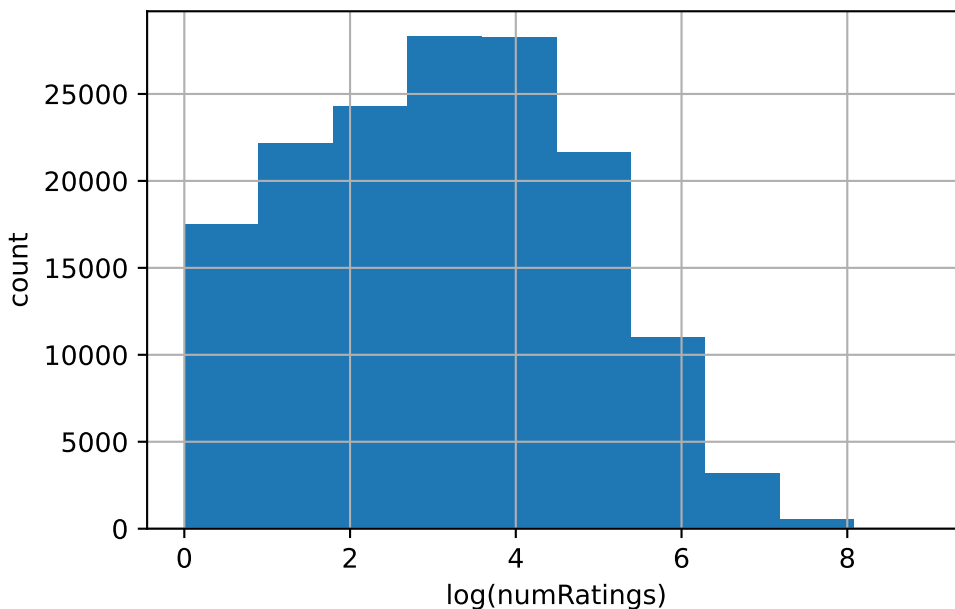
We fix the `cumsum=>2` issue, so we have authors who write 2 (not 3) notes as our outcome variable
We also remind the alignment filter, since this is conditioning on a collider (and makes our RDD artificially sharp)

```
Imported data for 3173276 notes
ALERT! re_auth_thresh = 2
Now we have 330234 first time note authors (all time)
Now we have 260997 first time note authors 2024-01-01 to 2026-06-02
Now we have 234819 notes written by our 4 choosen algorithms
Now we have 175194 notes classified as MISLEADING
Now we have 157044 notes which have >0 ratings
----- No filter for alignment
FINAL total 157044 Notes
of which author writes again = 107099
```

NB current (final), not Max, score

3 Diagnostic Plots

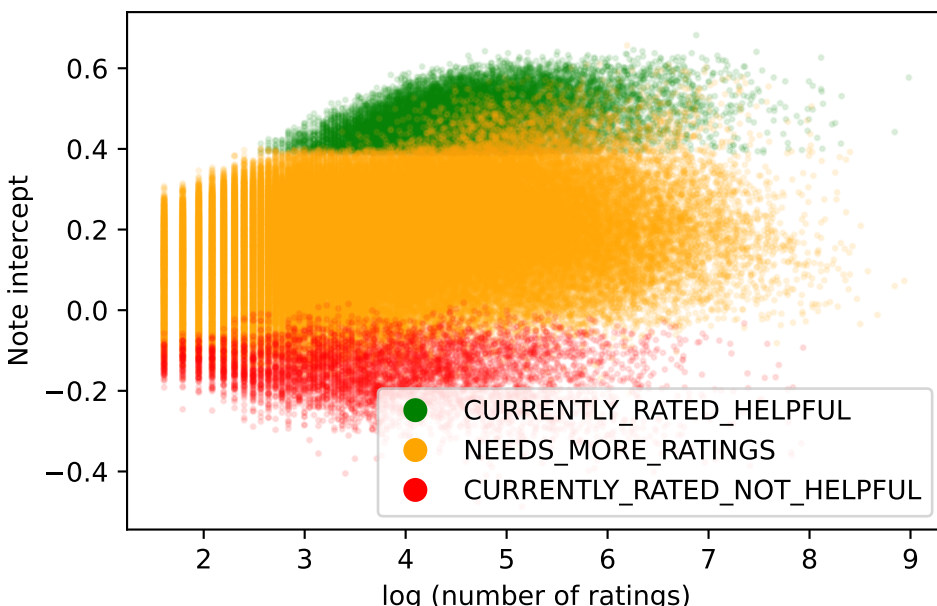
Most notes have a small number of rating, a small number have very many ratings This means the distribution of ratings is very skewed. We take the log of the number of ratings to make the distribution more normal



The basis for our RDD is the discontinuity between the intercept and note status (which defines publication / the possibility for publication)

Plotting $\log(\text{numRatings})$ vs NoteIntercept, with colours according to Note final status (finalRatingStatus)

You can see the fuzzy discontinuity - some >cutoff notes are “NEEDS_MORE_RATINGS”



Sadly, though, our data record intercept calculated from the full set of ratings, this is not the contemporaneous intercept score which determines moment-by-moment publication. It approximates it, but with contamination. Notes may be published then unpublished as they receive more ratings and their rating degrades, or there may be additional blocks on publication (e.g. requirements on total number or type of ratings, or requirements on intercept stability or a lag before publication). The CN system is not entirely clear about all these details.

These contamination can be seen by plotting indicators of ever having been published against our (final) intercept.

We construct an independent ‘ever_published’ variable from the `firstNonNMRStatus` and `mostRecentNonNMRStatus` variables. The logic being that if either of these are `CURRENTLY_RATED_HELPFUL` then the Note must have achieved CRH/published status at some point in the past (ie separate from the `finalRatingStatus` score)

```
[0.35,0.40): n= 4979 P(ever_pub)=0.3866 P(ever_crit)=0.4832
[0.38,0.40): n= 1942 P(ever_pub)=0.5695 P(ever_crit)=0.6560
[0.39,0.40): n= 1205 P(ever_pub)=0.7137 P(ever_crit)=0.7826
[0.40,0.42): n= 2102 P(ever_pub)=0.8782 P(ever_crit)=0.9172
```

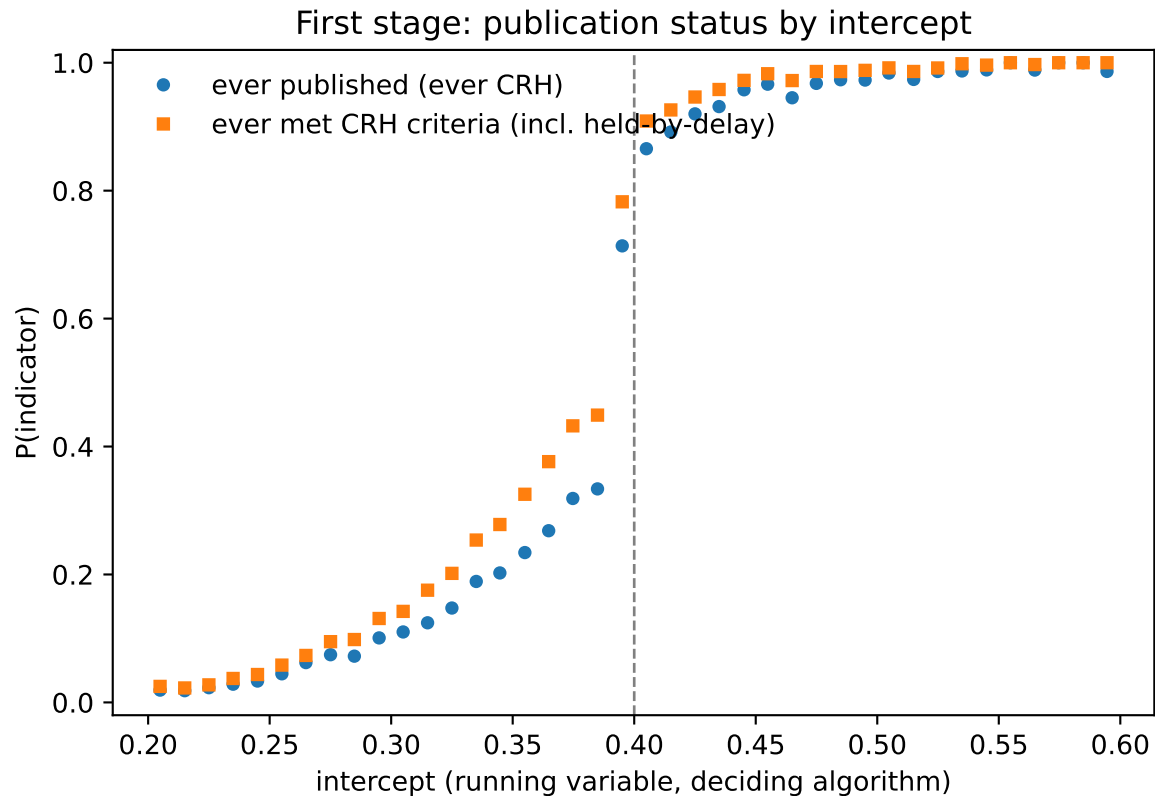
Below-cutoff `P(if_written_again)` by `ever_published`:

	mean	size
<code>ever_published</code>		
0	0.669529	104094
1	0.743547	4765

Below-cutoff `P(if_written_again)` by `ever_met_crh_criteria`:

	mean	size
<code>ever_met_crh_criteria</code>		

False	0.670453	102747
True	0.711715	6112



This shows that our secondary indicator of publication status shows a discontinuity near the cutoff, but not a sharp one

The high proportion of ‘ever published’ just below the cutoff suggests many notes got published and then slipped back below the threshold

We compromise with a fuzzy RDD

4 Trad OLS RDD

Our first approach was to use OLS and an arbitrary window

This was significant with the aligned notes, and a wide window (0.08, it was non-significant at 0.05), but now is not significant at the 0.08 window

bandwidth = 0.08

Number of treated units: 8504

Number of control units: 9624

Estimated Treatment Effect at the Cutoff: 0.0197

I think we can intepret this as a % change the chance of authoring again if published

P-value for treatment: 0.1699
T-statistic for treatment: 1.3725

5 RDD using fuzzy rdrobust with a donut

The rdrobust function will estimate the bandwidth from the data, as well as provide various sensitivity checks and allow a fuzzy RDD

It has the option of using a polynomial fit (which we decline), and uses all points, weighted by proximity to cutoff

We are using a ‘donut’ method of excluding those points near the cutoff, this is specifically to get those points just below the cutoff which have a high (outlier) percentage change of ever having been published

References for this method

@article{calonico2014robust, title = {Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {Econometrica}, volume = {82}, number = {6}, pages = {2295–2326}, year = {2014}, doi = {10.3982/ECTA11757} }

@article{calonico2018effect, title = {On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Farrell, Max H.}, journal = {Journal of the American Statistical Association}, volume = {113}, number = {522}, pages = {767–779}, year = {2018}, doi = {10.1080/01621459.2017.1285776} }

@article{calonico2014rdrobust, title = {Robust Data-Driven Inference in the Regression-Discontinuity Design}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {The Stata Journal}, volume = {14}, number = {4}, pages = {909–946}, year = {2014}, doi = {10.1177/1536867X1401400413} }

Justifying the donut RDD, we can test if intercept (our running variable) is associated with ‘ever published’

PRIMARY SPEC - donut ± 0.01

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	122098
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
Number of Observations	107654	14444
Number of Unique Obs.	107449	14426

Number of Effective Obs.	3630	4151
Bandwidth Estimation	0.049	0.049
Bandwidth Bias	0.119	0.119
rho (h/b)	0.409	0.409

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.486	0.031	15.549	1.626e-54	[0.424, 0.547]
Bias-Corrected	0.471	0.031	15.073	2.447e-51	[0.41, 0.532]
Robust	0.471	0.034	13.654	1.902e-42	[0.403, 0.538]

- - - - variation: donut ± 0.0

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	124389
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
Number of Observations	108859	15530
Number of Unique Obs.	108653	15510
Number of Effective Obs.	1864	1973
Bandwidth Estimation	0.019	0.019
Bandwidth Bias	0.058	0.058
rho (h/b)	0.325	0.325

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.014	0.027	0.516	6.059e-01	[-0.039, 0.066]
Bias-Corrected	0.0	0.027	0.002	9.986e-01	[-0.052, 0.053]
Robust	0.0	0.028	0.002	9.987e-01	[-0.054, 0.054]

- - - - variation: donut ± 0.005

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	123222
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd

```

Var-Cov Estimator:                      NN

                                Left      Right
-----
Number of Observations      108234      14988
Number of Unique Obs.      108028      14969
Number of Effective Obs.    1384       1646
Bandwidth Estimation        0.021     0.021
Bandwidth Bias              0.072     0.072
rho (h/b)                   0.291     0.291

```

```

Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional  -0.064    0.049    -1.292    1.964e-01    [-0.16, 0.033]
Bias-Corrected -0.09    0.049    -1.822    6.841e-02    [-0.186, 0.007]
Robust       -0.09    0.051    -1.755    7.922e-02    [-0.19, 0.01]

```

- - - - variation: donut ± 0.02

Call: rdrobust

Sharp RD estimates using local polynomial regression.

```

Number of Observations:      120345
Polynomial Order Est. (p):    1
Polynomial Order Bias (q):    2
Kernel:                      Triangular
Bandwidth Selection:          mserrd
Var-Cov Estimator:           NN

```

```

                                Left      Right
-----
Number of Observations      106917      13428
Number of Unique Obs.      106712      13412
Number of Effective Obs.    4909       4713
Bandwidth Estimation        0.063     0.063
Bandwidth Bias              0.179     0.179
rho (h/b)                   0.355     0.355

```

```

Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional  0.472    0.032    14.762    2.564e-49    [0.41, 0.535]
Bias-Corrected  0.45    0.032    14.046    8.199e-45    [0.387, 0.512]
Robust       0.45    0.034    13.097    3.440e-39    [0.382, 0.517]

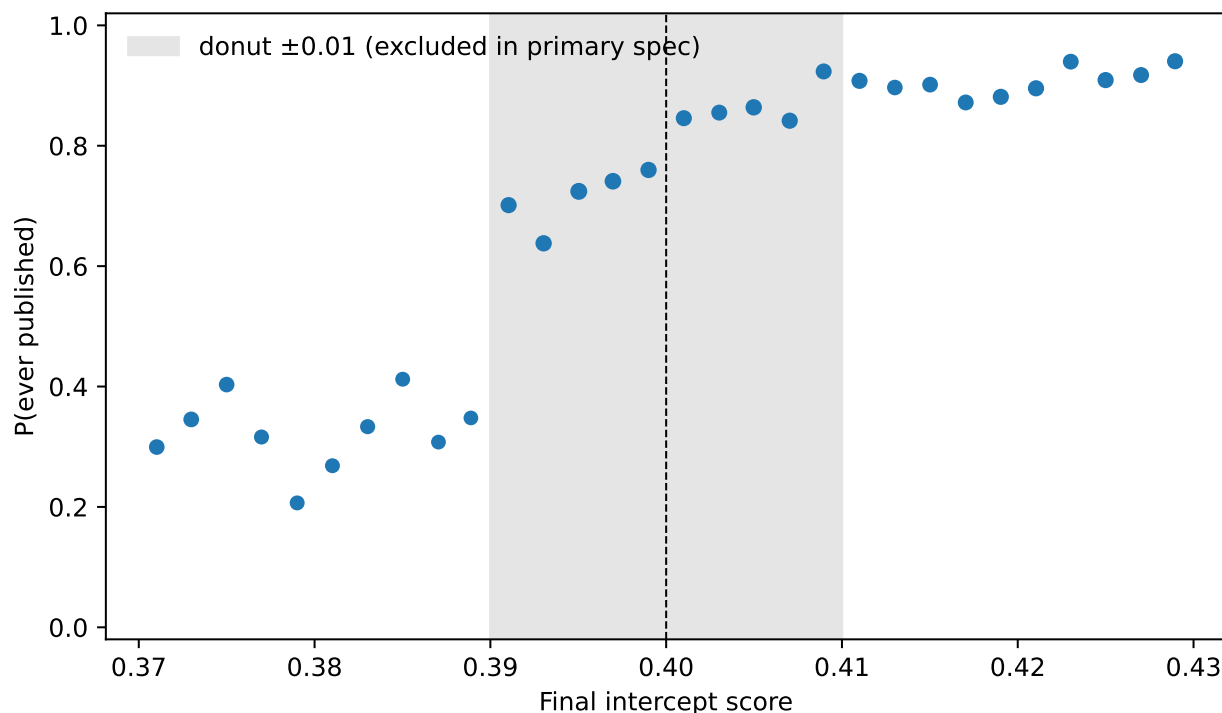
```

So this is saying that if you exclude at 0 (no exclusions) or 0.05 our running variable isn't cleanly associated with "ever published" (at 0.0 it is n.s. at 0.05 it is sig, but negative). At 0.01 the effect is positive and robust (significant), so this justifies the donut

The flip in significance shows significant contamination at the cutoff (those just below *more* likely

to be ever published than those just above!). So the argument is not just excluding problematic points, but the RDD is only possible against a background of linear stability

Note also the negligible drop in N with donut 0.01



'Mephi:\nWhy: the platform has a documented hysteresis rule of exactly 0.01. I just pulled the scoring/src/scoring/mf_base_scorer.py:178,188\nAddCRHInertia rule: a note that is already CRH 1 i.e. [0.39, 0.40). Class in scoring_rules.py:1521, docstring literally says "notes with interce hence the sharp break your plot shows precisely at the grey band\'s left edge.\n'

So this show the operation fo the inertiaDelta = 0.01 rule, see scoring_rules.py line 1521. Notes which are published and drop 0.01 or less keep their status <https://github.com/twitter/communitynotes/blob/>

nb OLS RDD now significant at 0.08 window with cutoff at 0.39, which is nice OR with a cutoff at 0.4 and a donut of 0.01

The OLS serves as a sanity check, but now let's run the full fuzzy RDD

```
running variable (x) length 122098
binary outcome (y) length 122098
cutoff (c) is 0.4
Call: rdrobust
```

Fuzzy RD estimates using local polynomial regression.

```
Number of Observations:      122098
Polynomial Order Est. (p):    1
Polynomial Order Bias (q):    2
Kernel:                      Triangular
```

Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	107654	14444
Number of Unique Obs.	107449	14426
Number of Effective Obs.	5256	5489
Bandwidth Estimation	0.061	0.061
Bandwidth Bias	0.147	0.147
rho (h/b)	0.416	0.416

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.076	0.059	1.284	1.990e-01	[-0.04, 0.192]
Robust	-	-	1.164	2.445e-01	[-0.052, 0.205]

First-Stage Estimates.

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.476	0.024	19.506	9.772e-85	[0.429, 0.524]
Robust	-	-	16.982	1.116e-64	[0.405, 0.511]

MSE-optimal h = 0.0610

Bias-corrected point estimate: 0.0765 95% CI [-0.0524, 0.2054]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0305 (0.5x) tau = +0.1352 95% CI [-0.1293, +0.4211] N_L/N_R = 1653/2148
h = 0.0610 (1.0x) tau = +0.0758 95% CI [-0.0524, +0.2054] N_L/N_R = 5256/5489
h = 0.1219 (2.0x) tau = +0.0841 95% CI [+0.0235, +0.1582] N_L/N_R = 17543/11177

6 Converting rate to baseline outcome rate

Common bandwidth h = 0.0610

Reduced form (ITT): +0.0361 robust CI [-0.0585, +0.1963]
First stage: +0.4765 robust CI [+0.3963, +0.6206]
Fuzzy LATE: +0.0758 robust CI [-0.0524, +0.2054]

Tie-out check: RF / FS = +0.0758 (should LATE at common h)

Mephi suggested text :

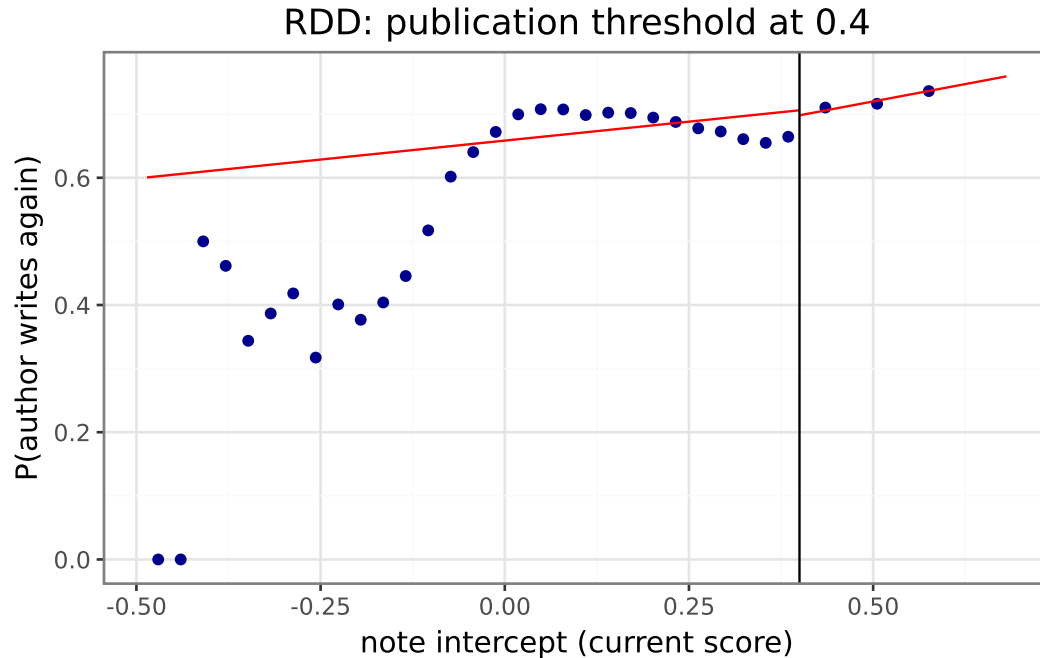
At the MSE-optimal bandwidth (h = 0.061) we estimate $\hat{\tau} = +7.6\text{pp}$ in the probability of writing a second note (conventional local-linear estimator; robust bias-corrected 95% CI [-5.2, +20.5], p = 0.24).

Mephi suggested text: Untreated authors at the cutoff re-author at 65.8%; treated authors at the

cutoff re-author at $\sim 73.4\%$. Treatment effect = 7.6pp, a 11.5% relative lift over the counterfactual baseline.

7 Showing an illustrative RDD plot

First, binned means with gloabl linear fit (this is the default plot from `rdplot`)



```
[ 'rdplot_mean_bin', 'rdplot_mean_x', 'rdplot_mean_y', 'rdplot_min_bin', 'rdplot_max_bin', 'rdplot_min_x', 'rdplot_max_x', 'rdplot_min_y', 'rdplot_max_y' ]
```

paper-outputs/RDDrobust_scatter.png + paper-outputs/RDDrobust_scatter_data.csv written (32 bins)

RD plot at $c = 0.4$ (p=1, triangular, N=122098)

```
bins used      : 29 left / 4 right
```

J_IMSE : [29.0, 4.0]

```
J_MV      : [974.0, 1000.0]
```

```
scale      : [1, 1]
```

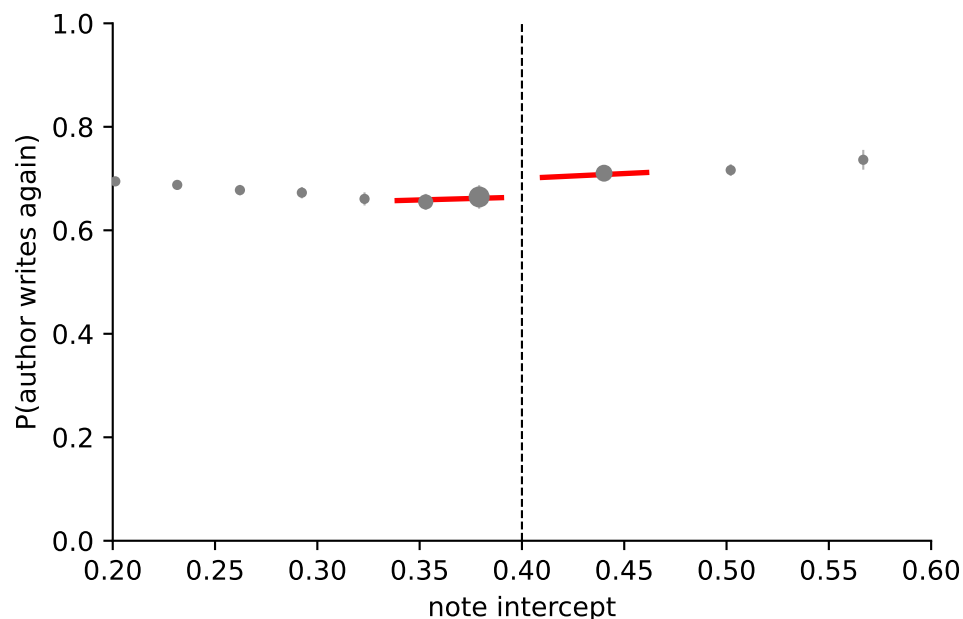
```
bin width      : avg [0.03054120705045503, 0.07023575305938719], median [0.030541207050455044, 0.05000000000000001]
```

coefficients:

	Left	Right
0	0.706061	0.698152
1	0.119146	0.218127

wrote paper-outputs/RDDrobust_scatter_meta.json, paper-outputs/RDDrobust_scatter_coef.csv

Now with the local-linear fits (over the `rdrobust` defined window)



consistency check: gap at cutoff should conventional tau
 plot gap at $c = +0.0361$ vs conventional tau = $+0.0758$

8 Robustness check - density test

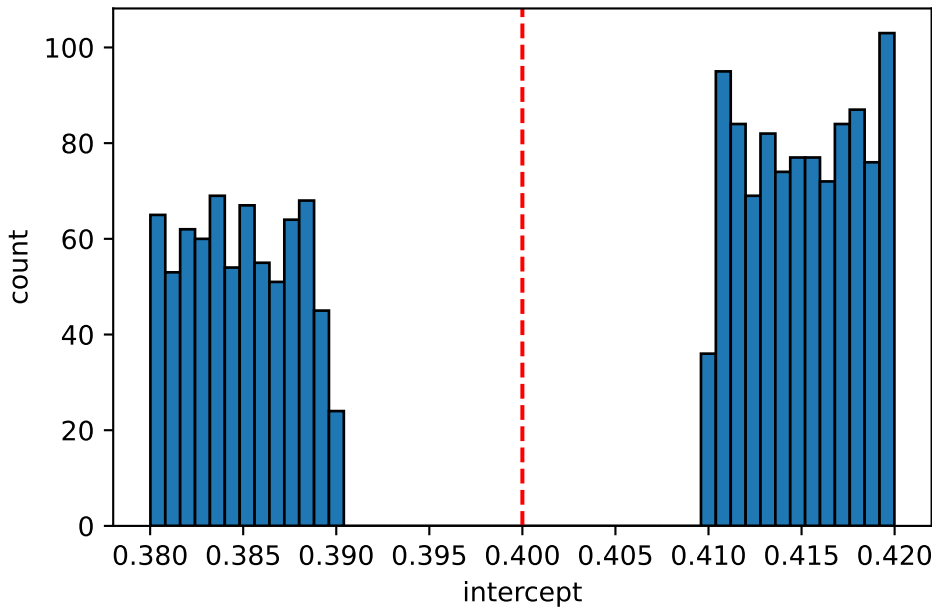
Next, manipulation test/density smoothness at cutoff

if units (people) can select to be just above cutoff then density will not be smooth (think the pass grade on an exam)

Inspecting density with a window of ± 0.02

```
t_asy      NaN
t_jk       0.945072
p_asy      NaN
p_jk       0.344622
dtype: float64
left       0.424248
right      0.577593
diff       0.153345
dtype: float64
left       0.057317
right      0.047540
dtype: float64
full       122098
left       107654
right      14444
eff_left   4704
eff_right  4001
dtype: int64
leftN      [737.0, 1050.0, 1451.0, 1827.0, 2229.0, 2686.0...
```

```
rightN      [1016.0, 1334.0, 1663.0, 2000.0, 2342.0, 2675....
leftWindow   [0.02, 0.02414638417454952, 0.0282927683490990...
rightWindow  [0.02, 0.023059989917801848, 0.026119979835603...
pval         [2.861901511174296e-11, 6.52541685793293e-09, ...
dtype: object
['X_max', 'X_min', 'binNW', 'bino', 'binoN', 'binoNStep', 'binoP', 'binoW', 'binoWStep', 'bino
```



A wider window shows that the density does progressively decrease up to the cutoff, but no signs of sub-threshold notes being shifted to above threshold (decline not matched by leap post-cutoff)

9 Robustness check - Donut RDD

Exluding results at the cutoff

donut	n_eff_L	n_eff_R	h	coef_conv	se_conv	p_robust	ci_robust_lo	ci_robust_hi
none	5256	5489	0.061	0.0758	0.0590	0.2445	-0.0524	0.2054
±0.005	5256	5489	0.061	0.0758	0.0590	0.2445	-0.0524	0.2054
±0.010	5256	5489	0.061	0.0758	0.0590	0.2445	-0.0524	0.2054
±0.020	5681	5220	0.068	-0.0024	0.0741	0.9352	-0.1715	0.1578

10 Sensitivity check - RDD sensitivity with different cutoffs

NB Only a cutoff of 0.4 makes sense in terms of the CN publication mechanism. Therefore we expect other thresholds to show now effect (definitely below 0.4, cutoffs above 0.4 will include some published notes above and below that cutoffs, so if there is a true effect may report an attenuated form of that)

```
running variable (x) length 122098
binary outcome (y) length 122098
cutoff (c) is 0.3
```

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 122098
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	95319	26779
Number of Unique Obs.	95139	26736
Number of Effective Obs.	15842	9832
Bandwidth Estimation	0.061	0.061
Bandwidth Bias	0.105	0.105
rho (h/b)	0.581	0.581

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.008	0.013	0.656	5.118e-01	[-0.017, 0.033]
Robust	-	-	0.744	4.570e-01	[-0.018, 0.04]

MSE-optimal h = 0.0609

Bias-corrected point estimate: 0.0111 95% CI [-0.0182, 0.0404]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0305 (0.5x) tau = +0.0234 95% CI [-0.0117, +0.0684] N_L/N_R = 7357/5728

h = 0.0609 (1.0x) tau = +0.0084 95% CI [-0.0182, +0.0404] N_L/N_R = 15842/9832

h = 0.1219 (2.0x) tau = +0.0019 95% CI [-0.0140, +0.0283] N_L/N_R = 35828/13554

running variable (x) length 122098

binary outcome (y) length 122098

cutoff (c) is 0.35

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 122098
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

Left	Right
------	-------

```

-----
Number of Observations      103880      18218
Number of Unique Obs.      103680      18195
Number of Effective Obs.    22592      8582
Bandwidth Estimation        0.105      0.105
Bandwidth Bias              0.159      0.159
rho (h/b)                   0.657      0.657

```

```

-----
Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional -0.003      0.013      -0.198      8.429e-01      [-0.028, 0.023]
Robust       -          -          -0.358      7.202e-01      [-0.035, 0.024]

```

MSE-optimal h = 0.1047

Bias-corrected point estimate: -0.0054 95% CI [-0.0351, 0.0242]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

```

h = 0.0523 (0.5x) tau = +0.0074 95% CI [-0.0328, +0.0548] N_L/N_R = 9084/3774
h = 0.1047 (1.0x) tau = -0.0025 95% CI [-0.0351, +0.0242] N_L/N_R = 22592/8582
h = 0.2093 (2.0x) tau = +0.0041 95% CI [-0.0205, +0.0251] N_L/N_R = 57002/17097

```

running variable (x) length 14444

binary outcome (y) length 14444

cutoff (c) is 0.45

Call: rdrobust

Sharp RD estimates using local polynomial regression.

```

Number of Observations:      14444
Polynomial Order Est. (p):      1
Polynomial Order Bias (q):      2
Kernel:                        Triangular
Bandwidth Selection:            mserd
Var-Cov Estimator:              NN

```

```

-----
Left      Right
-----
Number of Observations      4289      10155
Number of Unique Obs.      4284      10142
Number of Effective Obs.    1056      1073
Bandwidth Estimation        0.01      0.01
Bandwidth Bias              0.017      0.017
rho (h/b)                   0.565      0.565

```

```

-----
Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional 0.088      0.043      2.058      3.960e-02      [0.004, 0.173]
Robust       -          -          2.121      3.391e-02      [0.008, 0.201]

```

MSE-optimal $h = 0.0097$

Bias-corrected point estimate: 0.1045 95% CI [0.0079, 0.2010]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0048 (0.5x) tau = +0.0579 95% CI [-0.0901, +0.1860] N_L/N_R = 555/539
h = 0.0097 (1.0x) tau = +0.0884 95% CI [+0.0079, +0.2010] N_L/N_R = 1056/1073
h = 0.0193 (2.0x) tau = +0.0429 95% CI [-0.0218, +0.1136] N_L/N_R = 2123/2036

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 14444
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	4289	10155
Number of Unique Obs.	4284	10142
Number of Effective Obs.	1056	1073
Bandwidth Estimation	0.01	0.01
Bandwidth Bias	0.017	0.017
rho (h/b)	0.565	0.565

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.088	0.043	2.058	3.960e-02	[0.004, 0.173]
Robust	-	-	2.121	3.391e-02	[0.008, 0.201]

All these are non-significant

11 Sensitivity check - data for different time windows

PERIOD 1, from 2025-04-08 to 2026-06-01

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 87791 first time note authors 2025-04-08 to 2026-06-01

Now we have 80008 notes written by our 4 choosen algorithms

Now we have 62447 notes classified as MISLEADING

Now we have 55994 notes which have >0 ratings

----- No filter for alignment

FINAL total 55994 Notes

of which author writes again = 31560

NB current (final), not Max, score
running variable (x) length 55994
binary outcome (y) length 55994
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 43770
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	38117	5653
Number of Unique Obs.	38085	5646
Number of Effective Obs.	2845	2846
Bandwidth Estimation	0.066	0.066
Bandwidth Bias	0.102	0.102
rho (h/b)	0.653	0.653

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.004	0.029	0.153	8.784e-01	[-0.052, 0.061]
Robust	-	-	-0.065	9.484e-01	[-0.069, 0.065]

MSE-optimal h = 0.0664

Bias-corrected point estimate: -0.0022 95% CI [-0.0692, 0.0648]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0332 (0.5x) tau = +0.0006 95% CI [-0.0934, +0.0951] N_L/N_R = 1196/1442
h = 0.0664 (1.0x) tau = +0.0044 95% CI [-0.0692, +0.0648] N_L/N_R = 2845/2846
h = 0.1329 (2.0x) tau = +0.0261 95% CI [-0.0317, +0.0658] N_L/N_R = 8054/4924

PERIOD 1, from 2024-01-01 to 2025-04-07

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 172844 first time note authors 2024-01-01 to 2025-04-07

Now we have 154486 notes written by our 4 choosen algorithms

Now we have 112502 notes classified as MISLEADING

Now we have 100832 notes which have >0 ratings

----- No filter for alignment

FINAL total 100832 Notes

of which author writes again = 75433

NB current (final), not Max, score
running variable (x) length 100832
binary outcome (y) length 100832
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 80440
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	70580	9860
Number of Unique Obs.	70473	9858
Number of Effective Obs.	3823	3877
Bandwidth Estimation	0.06	0.06
Bandwidth Bias	0.101	0.101
rho (h/b)	0.59	0.59

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.001	0.021	-0.048	9.617e-01	[-0.043, 0.041]
Robust	-	-	-0.362	7.170e-01	[-0.057, 0.039]

MSE-optimal h = 0.0599

Bias-corrected point estimate: -0.0089 95% CI [-0.0567, 0.0390]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0299 (0.5x)	tau = -0.0195	95% CI [-0.0936, +0.0410]	N_L/N_R = 1721/1869
h = 0.0599 (1.0x)	tau = -0.0010	95% CI [-0.0567, +0.0390]	N_L/N_R = 3823/3877
h = 0.1198 (2.0x)	tau = +0.0196	95% CI [-0.0203, +0.0494]	N_L/N_R = 11285/7472